

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Integrals

1.1 Medium Integral

Trigonometric-Exponential Improper Integral

Evaluate the definite integral:

$$\int_0^{\infty} \frac{\sin^2(x) \cos(x)}{e^x} dx$$

Solution: Let the integral be denoted by I :

$$I = \int_0^{\infty} e^{-x} \sin^2(x) \cos(x) dx$$

First, we linearize the trigonometric product $\sin^2(x) \cos(x)$ to express it as a sum of fundamental harmonics. Using the power-reduction identity $\sin^2(x) = \frac{1 - \cos(2x)}{2}$, we obtain:

$$\sin^2(x) \cos(x) = \left(\frac{1 - \cos(2x)}{2} \right) \cos(x) = \frac{1}{2} \cos(x) - \frac{1}{2} \cos(2x) \cos(x)$$

Applying the product-to-sum identity $\cos(A) \cos(B) = \frac{1}{2} [\cos(A + B) + \cos(A - B)]$ for the second term:

$$\frac{1}{2} \cos(2x) \cos(x) = \frac{1}{4} [\cos(3x) + \cos(x)]$$

Substituting this back yields the simplified trigonometric integrand:

$$\sin^2(x) \cos(x) = \frac{1}{2} \cos(x) - \frac{1}{4} \cos(3x) - \frac{1}{4} \cos(x) = \frac{1}{4} \cos(x) - \frac{1}{4} \cos(3x)$$

Now, we substitute this linear combination into the original integral I :

$$I = \frac{1}{4} \int_0^{\infty} e^{-x} \cos(x) dx - \frac{1}{4} \int_0^{\infty} e^{-x} \cos(3x) dx$$

Recall the standard Laplace transform of cosine, or equivalently the standard integration formula:

$$\int_0^{\infty} e^{-ax} \cos(bx) dx = \frac{a}{a^2 + b^2} \quad \text{for } a > 0$$

Applying this formula to our two integrals with $a = 1$ and $b \in \{1, 3\}$:

- For $b = 1$: $\int_0^{\infty} e^{-x} \cos(x) dx = \frac{1}{1^2 + 1^2} = \frac{1}{2}$
- For $b = 3$: $\int_0^{\infty} e^{-x} \cos(3x) dx = \frac{1}{1^2 + 3^2} = \frac{1}{10}$

Substituting these values back into our expression for I :

$$I = \frac{1}{4} \left(\frac{1}{2} \right) - \frac{1}{4} \left(\frac{1}{10} \right) = \frac{1}{8} - \frac{1}{40}$$

Finding a common denominator of 40:

$$I = \frac{5}{40} - \frac{1}{40} = \frac{4}{40} = \frac{1}{10}$$

The exact value is:

$$\frac{1}{10}$$

1.2 Hard Integral

Logarithmic-Algebraic Integral via Beta Function

Evaluate the definite integral:

$$\int_0^1 \frac{\ln(x) \ln(1-x)}{\sqrt{x(1-x)}} dx$$

Solution: Let the target integral be denoted by I . We recognize the integrand is closely related to the Eulerian Beta function. The integral can be expressed as:

$$I = \int_0^1 x^{-1/2} (1-x)^{-1/2} \ln(x) \ln(1-x) dx$$

Recall the integral definition of the Beta function:

$$B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

Differentiating $B(\alpha, \beta)$ with respect to α brings down a factor of $\ln(x)$, and differentiating with respect to β brings down a factor of $\ln(1-x)$. Therefore, our integral I is precisely the mixed second-order partial derivative of the Beta function evaluated at $\alpha = 1/2$ and $\beta = 1/2$:

$$I = \frac{\partial^2}{\partial \alpha \partial \beta} B(\alpha, \beta) \Big|_{\alpha=1/2, \beta=1/2}$$

We express the first derivative of the Beta function using the digamma function $\psi(z) = \frac{d}{dz} \ln \Gamma(z) = \frac{\Gamma'(z)}{\Gamma(z)}$:

$$\frac{\partial}{\partial \alpha} B(\alpha, \beta) = B(\alpha, \beta) [\psi(\alpha) - \psi(\alpha + \beta)]$$

Taking the derivative with respect to β :

$$\frac{\partial^2}{\partial \alpha \partial \beta} B(\alpha, \beta) = \frac{\partial}{\partial \beta} \{B(\alpha, \beta) [\psi(\alpha) - \psi(\alpha + \beta)]\}$$

Using the product rule and chain rule:

$$\frac{\partial^2 B}{\partial \alpha \partial \beta} = B(\alpha, \beta) [\psi(\alpha) - \psi(\alpha + \beta)] [\psi(\beta) - \psi(\alpha + \beta)] - B(\alpha, \beta) \psi'(\alpha + \beta)$$

Factoring out $B(\alpha, \beta)$, we obtain the explicit general formula:

$$\frac{\partial^2 B}{\partial \alpha \partial \beta} = B(\alpha, \beta) \{ [\psi(\alpha) - \psi(\alpha + \beta)] [\psi(\beta) - \psi(\alpha + \beta)] - \psi'(\alpha + \beta) \}$$

Now we evaluate this expression at $\alpha = 1/2$ and $\beta = 1/2$:

- $B(1/2, 1/2) = \frac{\Gamma(1/2)\Gamma(1/2)}{\Gamma(1)} = \frac{\sqrt{\pi} \cdot \sqrt{\pi}}{1} = \pi$
- Sum of arguments: $\alpha + \beta = 1$. The digamma values are $\psi(1) = -\gamma$ and $\psi(1/2) = -\gamma - 2\ln(2)$.
- Difference terms: $\psi(1/2) - \psi(1) = (-\gamma - 2\ln(2)) - (-\gamma) = -2\ln(2)$.
- Trigamma value: $\psi'(1) = \sum_{k=1}^{\infty} \frac{1}{k^2} = \zeta(2) = \frac{\pi^2}{6}$.

Substituting these specific values into the derivative formula:

$$I = \pi \left\{ [-2\ln(2)] [-2\ln(2)] - \frac{\pi^2}{6} \right\}$$

$$I = \pi \left(4\ln^2(2) - \frac{\pi^2}{6} \right)$$

The exact analytical value is:

$$\pi \left(4\ln^2(2) - \frac{\pi^2}{6} \right)$$

Differentials

2.1 Medium Differential

Fifth-Order Linear Differential Equation with Limit Constraints

Solve the differential equation

$$\frac{d^5 y}{dx^5} - \frac{d^4 y}{dx^4} + 2\frac{d^3 y}{dx^3} - 2\frac{d^2 y}{dx^2} + \frac{dy}{dx} - y = 0$$

Knowing that $y(0) = 0$, $y'(0) > 0$, $\liminf_{x \rightarrow \infty} y(x) = -2$ and $\limsup_{x \rightarrow \infty} y(x) = 2$, find $y(2026)$.

Solution: We are given a fifth-order homogeneous linear ordinary differential equation with constant coefficients. We start by writing its characteristic equation:

$$r^5 - r^4 + 2r^3 - 2r^2 + r - 1 = 0$$

We can solve this by factoring by grouping:

$$r^4(r - 1) + 2r^2(r - 1) + 1(r - 1) = 0$$

Factoring out the common term $(r - 1)$:

$$(r - 1)(r^4 + 2r^2 + 1) = 0$$

Notice that the second polynomial is a perfect square:

$$(r - 1)(r^2 + 1)^2 = 0$$

The roots of the characteristic equation are $r_1 = 1$, and double complex roots $r_2, r_3 = i$ and $r_4, r_5 = -i$. Thus, the general solution to the differential equation is a linear combination of the corresponding basis functions:

$$y(x) = C_1 e^x + (C_2 + C_3 x) \cos(x) + (C_4 + C_5 x) \sin(x)$$

We now apply the asymptotic behavior conditions as $x \rightarrow \infty$. The problem states that $\liminf_{x \rightarrow \infty} y(x) = -2$ and $\limsup_{x \rightarrow \infty} y(x) = 2$. Since these limits are finite, any unbounded terms in the general solution must vanish. Specifically:

- The exponential term e^x grows without bound, so we must have $C_1 = 0$.
- The secular terms $x \cos(x)$ and $x \sin(x)$ have linearly growing amplitudes, so we must have $C_3 = 0$ and $C_5 = 0$.

This vastly simplifies our solution to a purely oscillatory function:

$$y(x) = C_2 \cos(x) + C_4 \sin(x)$$

This function can be rewritten as $y(x) = A \sin(x + \phi)$, where the amplitude is $A = \sqrt{C_2^2 + C_4^2}$. The lim sup and lim inf constraints mandate that the amplitude of this sinusoidal wave is exactly 2. Thus:

$$\sqrt{C_2^2 + C_4^2} = 2$$

Next, we apply the initial conditions at $x = 0$:

$$y(0) = C_2 \cos(0) + C_4 \sin(0) = C_2$$

Since we are given $y(0) = 0$, it follows that $C_2 = 0$. Substituting $C_2 = 0$ into the amplitude equation gives:

$$\sqrt{0^2 + C_4^2} = 2 \implies |C_4| = 2 \implies C_4 = \pm 2$$

To determine the correct sign, we differentiate our simplified $y(x)$:

$$y(x) = C_4 \sin(x) \implies y'(x) = C_4 \cos(x)$$

Using the derivative initial condition $y'(0) > 0$:

$$y'(0) = C_4 \cos(0) = C_4 > 0 \implies C_4 = 2$$

Therefore, the particular solution to the differential equation is:

$$y(x) = 2 \sin(x)$$

Finally, to find $y(2026)$, we evaluate the particular solution at $x = 2026$:

$$y(2026) = 2 \sin(2026)$$

The exact evaluated expression is:

$$2 \sin(2026)$$

2.2 Hard Differential

Integro-Differential Equation and Integral Evaluation

Given the continuous and differentiable function $y : [0, \infty) \rightarrow \mathbb{R}$ that satisfies the following integro-differential equation:

$$y'(x) = y(x) + \frac{1}{x} \int_0^x y(t) dt \quad \text{for } x > 0$$

given the initial condition $\lim_{x \rightarrow 0^+} y(x) = 1$. Evaluate the definite integral:

$$I = \int_0^1 y(x) dx$$

Solution: We are tasked with solving an integro-differential equation. We start by multiplying both sides by x to clear the fractional term and isolate the definite integral:

$$xy'(x) = xy(x) + \int_0^x y(t) dt$$

To eliminate the integral, we differentiate the entire equation with respect to x . By applying the product rule on the left and the Fundamental Theorem of Calculus on the right, we obtain:

$$\frac{d}{dx}[xy'(x)] = \frac{d}{dx}[xy(x)] + \frac{d}{dx} \left[\int_0^x y(t) dt \right]$$

$$y'(x) + xy''(x) = y(x) + xy'(x) + y(x)$$

We can rearrange this second-order differential equation into standard form by collecting like terms:

$$xy''(x) + (1-x)y'(x) - 2y(x) = 0$$

This equation is a special case of Kummer's equation, $xy'' + (c-x)y' - ay = 0$, with parameters $a = 2$ and $c = 1$. Rather than using generalized hypergeometric functions, we can search for a closed-form solution. Let us assume a solution of the form $y(x) = u(x)e^x$. Differentiating this ansatz:

$$y'(x) = u'(x)e^x + u(x)e^x = (u' + u)e^x$$

$$y''(x) = u''(x)e^x + 2u'(x)e^x + u(x)e^x = (u'' + 2u' + u)e^x$$

Substitute these derivatives into our differential equation:

$$x(u'' + 2u' + u)e^x + (1-x)(u' + u)e^x - 2ue^x = 0$$

Divide by $e^x \neq 0$ and expand:

$$xu'' + 2xu' + xu + u' + u - xu' - xu - 2u = 0$$

$$xu'' + (x+1)u' - u = 0$$

Notice that $u(x) = 1+x$ is a simple polynomial solution to this equation, as $u' = 1$ and $u'' = 0$:

$$x(0) + (x+1)(1) - (1+x) = 0$$

Thus, our solution candidate is $y(x) = C(1+x)e^x$. Let us verify the initial condition $\lim_{x \rightarrow 0^+} y(x) = 1$:

$$\lim_{x \rightarrow 0^+} C(1+x)e^x = C(1)(1) = C$$

Therefore, $C = 1$, and our unique function is $y(x) = (1+x)e^x$. Let us rapidly verify it satisfies the original integro-differential relation:

$$\int_0^x (1+t)e^t dt = [te^t]_0^x = xe^x$$

$$y(x) + \frac{1}{x} \int_0^x y(t) dt = (1+x)e^x + \frac{1}{x}(xe^x) = (2+x)e^x$$

$$y'(x) = \frac{d}{dx}[(1+x)e^x] = e^x + (1+x)e^x = (2+x)e^x$$

The function exactly satisfies the equation. We are tasked with evaluating $I = \int_0^1 y(x) dx$:

$$I = \int_0^1 (1+x)e^x dx = [xe^x]_0^1 = (1)e^1 - (0)e^0 = e$$

The exact evaluation of the definite integral is:

$$e$$

Derivatives

3.1 Medium Derivative

Absolute Cotangent Integral Derivative

Let $f(x) = \int_0^x \ln \left(\sqrt{|\cot(\frac{t}{2})|} \right) dt$. Find $f^{(3)}(\frac{7\pi}{12})$.

Solution: We apply the First Fundamental Theorem of Calculus to find the first derivative of $f(x)$:

$$f'(x) = \ln \left(\sqrt{|\cot(\frac{x}{2})|} \right)$$

Using the logarithmic property $\ln(a^b) = b \ln(a)$, we extract the square root:

$$f'(x) = \frac{1}{2} \ln |\cot(\frac{x}{2})|$$

We evaluate the target point $x = \frac{7\pi}{12}$. Here, $\frac{x}{2} = \frac{7\pi}{24}$, which lies in the first quadrant, so $\cot(\frac{x}{2}) > 0$ and the absolute value is trivially dropped for differentiation in the local neighborhood. We compute the second derivative $f''(x)$ using the chain rule:

$$f''(x) = \frac{1}{2} \cdot \frac{1}{\cot(\frac{x}{2})} \cdot \frac{d}{dx} \left[\cot\left(\frac{x}{2}\right) \right]$$

$$f''(x) = \frac{1}{2} \cdot \tan\left(\frac{x}{2}\right) \cdot \left[-\frac{1}{2} \csc^2\left(\frac{x}{2}\right) \right]$$

By expressing tangent and cosecant in terms of sine and cosine:

$$f''(x) = -\frac{1}{4} \left(\frac{\sin(x/2)}{\cos(x/2)} \right) \left(\frac{1}{\sin^2(x/2)} \right) = -\frac{1}{4 \sin(x/2) \cos(x/2)}$$

Applying the double angle identity $\sin(x) = 2 \sin(x/2) \cos(x/2)$:

$$f''(x) = -\frac{1}{2 \sin(x)} = -\frac{1}{2} \csc(x)$$

Now we take the third derivative $f^{(3)}(x)$ by differentiating $-\frac{1}{2} \csc(x)$:

$$f^{(3)}(x) = -\frac{1}{2} [-\csc(x) \cot(x)] = \frac{1}{2} \csc(x) \cot(x)$$

We evaluate this at $x = \frac{7\pi}{12}$. This angle corresponds to 105° . We can find the exact trigonometric values using angle addition $105^\circ = 60^\circ + 45^\circ$. For sine:

$$\sin(105^\circ) = \sin(60^\circ) \cos(45^\circ) + \cos(60^\circ) \sin(45^\circ) = \left(\frac{\sqrt{3}}{2} \right) \left(\frac{\sqrt{2}}{2} \right) + \left(\frac{1}{2} \right) \left(\frac{\sqrt{2}}{2} \right) = \frac{\sqrt{6} + \sqrt{2}}{4}$$

Taking the reciprocal gives the cosecant:

$$\csc(105^\circ) = \frac{4}{\sqrt{6} + \sqrt{2}}$$

Rationalizing the denominator:

$$\csc(105^\circ) = \frac{4(\sqrt{6} - \sqrt{2})}{6 - 2} = \sqrt{6} - \sqrt{2}$$

For cosine:

$$\cos(105^\circ) = \cos(60^\circ)\cos(45^\circ) - \sin(60^\circ)\sin(45^\circ) = \frac{\sqrt{2} - \sqrt{6}}{4}$$

Thus, the cotangent is the ratio of cosine to sine:

$$\cot(105^\circ) = \frac{\cos(105^\circ)}{\sin(105^\circ)} = \frac{\sqrt{2} - \sqrt{6}}{\sqrt{6} + \sqrt{2}}$$

Rationalizing the denominator:

$$\cot(105^\circ) = \frac{(\sqrt{2} - \sqrt{6})^2}{6 - 2} = \frac{2 - 2\sqrt{12} + 6}{-4} = \frac{8 - 4\sqrt{3}}{-4} = -2 + \sqrt{3}$$

Substitute these exact values back into our third derivative expression:

$$f^{(3)}\left(\frac{7\pi}{12}\right) = \frac{1}{2}(\sqrt{6} - \sqrt{2})(-2 + \sqrt{3})$$

Expanding the product:

$$(\sqrt{6} - \sqrt{2})(\sqrt{3} - 2) = \sqrt{18} - 2\sqrt{6} - \sqrt{6} + 2\sqrt{2} = 3\sqrt{2} - 3\sqrt{6} + 2\sqrt{2} = 5\sqrt{2} - 3\sqrt{6}$$

Multiplying by the factor of $\frac{1}{2}$:

$$f^{(3)}\left(\frac{7\pi}{12}\right) = \frac{5\sqrt{2} - 3\sqrt{6}}{2}$$

3.2 Hard Derivative

Maclaurin Series Expansion Evaluation

Let $f(x) = \tan^5(x)$. Find $f^{(5)}(0)$.

Solution: We are tasked to evaluate the fifth derivative of the function $f(x) = \tan^5(x)$ exactly at the origin $x = 0$. Using the chain and product rules directly for five successive derivations becomes algebraically tedious. Instead, we appeal to the definition of the Maclaurin series expansion of a sufficiently smooth function:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(5)}(0)}{5!}x^5 + \cdots$$

The value of the n -th derivative evaluated at $x = 0$ is directly proportional to the coefficient of the x^n term in the Maclaurin series of $f(x)$. Specifically:

$$f^{(5)}(0) = 5! \cdot [x^5] \text{ coefficient of } f(x)$$

Let us construct the series expansion for $f(x) = \tan^5(x)$ around $x = 0$. The well-known Maclaurin series expansion for the standard tangent function is:

$$\tan(x) = x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \mathcal{O}(x^7)$$

Since we need the expansion for $\tan^5(x)$, we raise this series to the fifth power:

$$\tan^5(x) = \left(x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \mathcal{O}(x^7) \right)^5$$

Factoring out the dominant linear term x :

$$\tan^5(x) = x^5 \left(1 + \frac{1}{3}x^2 + \frac{2}{15}x^4 + \mathcal{O}(x^6) \right)^5$$

We are only interested in finding the coefficient of the x^5 term in this expansion. By expanding the term in the parenthesis using the binomial theorem (or observing the lowest degree term), we see that the leading term of the entire expansion is strictly x^5 :

$$\tan^5(x) = x^5 \left(1^5 + \binom{5}{1} 1^4 \left(\frac{1}{3}x^2 \right) + \dots \right) = x^5 \left(1 + \frac{5}{3}x^2 + \dots \right) = x^5 + \frac{5}{3}x^7 + \dots$$

Therefore, the coefficient of x^5 is exactly 1. Equating this to our Maclaurin series coefficient formula:

$$\frac{f^{(5)}(0)}{5!} = 1$$

Solving for $f^{(5)}(0)$:

$$f^{(5)}(0) = 5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

The exact value is:

$$120$$

Physics & Engineering

4.1 Mechanics

Coupled Oscillator Angular Frequency

A light, inextensible string passes over a smooth, frictionless pulley. One end is attached to a bob of mass $m_1 = 4 \text{ kg}$ connected to a vertical spring of constant $k = 60 \text{ N m}^{-1}$ fixed to the ground. The other end is attached to a mass $m_2 = 11 \text{ kg}$ which hangs freely. The system is released from rest. Find the angular frequency ω (in rad s^{-1}). Take $g = 9.81 \text{ m s}^{-2}$.

Solution: Let us define the coordinate x as the downward displacement of the mass m_2 from the static equilibrium position of the system. Because the string is inextensible, a downward displacement x of m_2 corresponds exactly to an upward displacement x of mass m_1 . We can write the total mechanical energy of the system E as a function of the displacement x and the velocity $\dot{x}$. The kinetic energy T of the entire system consists of the motion of both masses:

$$T = \frac{1}{2}m_1\dot{x}^2 + \frac{1}{2}m_2\dot{x}^2 = \frac{1}{2}(m_1 + m_2)\dot{x}^2$$

The potential energy U relative to the equilibrium position involves both the change in gravitational potential and the elastic potential energy stored in the spring. However, because x is measured from the equilibrium position, the linear terms corresponding to gravitational potential energy and the initial spring stretch exactly cancel out. The potential energy is simply governed by the harmonic restoring force of the spring:

$$U = \frac{1}{2}kx^2 + \text{constant}$$

The total mechanical energy $E = T + U$ is conserved since all non-conservative forces do zero net work (frictionless pulley, ideal spring). Taking the time derivative of E gives zero:

$$\begin{aligned} \frac{dE}{dt} &= \frac{d}{dt} \left[\frac{1}{2}(m_1 + m_2)\dot{x}^2 + \frac{1}{2}kx^2 \right] = 0 \\ (m_1 + m_2)\dot{x}\ddot{x} + kx\dot{x} &= 0 \end{aligned}$$

For non-trivial motion ($\dot{x} \neq 0$), we divide by $\dot{x}$:

$$(m_1 + m_2)\ddot{x} + kx = 0 \implies \ddot{x} = -\frac{k}{m_1 + m_2}x$$

This defines a classic Simple Harmonic Oscillator (SHO) differential equation $\ddot{x} = -\omega^2 x$. The square of the angular frequency is defined as the ratio of the effective stiffness to the effective inertia:

$$\omega^2 = \frac{k}{m_1 + m_2}$$

Substituting the given system parameters, $k = 60 \text{ N m}^{-1}$, $m_1 = 4 \text{ kg}$, and $m_2 = 11 \text{ kg}$:

$$\omega = \sqrt{\frac{60}{4 + 11}} = \sqrt{\frac{60}{15}} = \sqrt{4} = 2 \text{ rad s}^{-1}$$

Notice that the acceleration due to gravity g affects the static equilibrium position but has no impact on the resulting angular frequency of oscillation. The angular frequency of the system is:

$$2 \text{ rad s}^{-1}$$

4.2 Quantum Mechanics

Nuclear Reactor Neutron Population Multiplication

A thermal nuclear reactor has an effective neutron multiplication factor of $k = 0.9965$ after its control rods are inserted. Immediately after insertion, the reactor contains $N_0 = 2.50 \times 10^{11}$ neutrons in one neutron generation. After $n = 400$ generations calculate the neutron population. Give your answer in the form $A \times 10^{10}$ and provide the value of A to three significant figures.

Solution: The neutron population in a thermal nuclear reactor evolves according to a geometric progression model governed by the effective neutron multiplication factor k . If N_0 is the initial population at generation zero, the population N_n after n successive generations is mathematically defined as:

$$N_n = N_0 k^n$$

We are given the initial population $N_0 = 2.50 \times 10^{11}$, a subcritical multiplication factor $k = 0.9965$, and the number of generations $n = 400$. We plug these parameters into our general equation:

$$N_{400} = (2.50 \times 10^{11}) \times (0.9965)^{400}$$

To find $(0.9965)^{400}$ with sufficient analytical precision, we compute the exponentiation:

$$(0.9965)^{400} \approx 0.2458925$$

Substituting this decay ratio back into the population expression:

$$N_{400} = (2.50 \times 10^{11}) \times 0.2458925 = 0.614731 \times 10^{11}$$

We are asked to format the final answer in scientific notation in the specific form $A \times 10^{10}$. To achieve this, we shift the decimal point one place to the right:

$$0.614731 \times 10^{11} = 6.14731 \times 10^{10}$$

Therefore, the pre-factor A is 6.14731. The problem requires us to state the value of A rounded to three significant figures. Looking at the fourth digit (7), we round up the third digit:

$$A \approx 6.15$$

The value of A to three significant figures is:

$$6.15$$

4.3 Electromagnetism

Faraday's Laws of Electrolysis for Series Cells

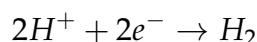
Two electrolytic cells are connected in series. The first cell contains an aqueous solution of H_2SO_4 and the second contains an aqueous solution of $CuSO_4$. During electrolysis with a constant current 10^{-5} kg of hydrogen is released in one minute. Determine the mass of copper deposited in the second cell during four minutes of electrolysis.

Solution: Because the two electrolytic cells are electrically connected in series, the identical constant current I passes through both of them. Consequently, the same quantity of electric charge $Q = It$ flows through both cells for any given duration t . We can use Faraday's Laws of Electrolysis to analyze the chemical reactions.

According to Faraday's First Law, the mass m of a substance deposited or liberated at an electrode is directly proportional to the total charge Q . It can be expressed via the molar mass M and the valency (number of electrons transferred per ion) z as:

$$m = \frac{Q}{F} \frac{M}{z} \implies Q = \frac{mzF}{M}$$

where F is Faraday's constant. In the first cell, aqueous H_2SO_4 is electrolyzed, liberating hydrogen gas (H_2) at the cathode. The corresponding cathodic half-reaction is:



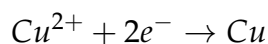
For hydrogen gas, the valency factor $z = 2$ electrons per mole of H_2 , and the molar mass is $M_{H_2} = 2.0 \text{ g mol}^{-1} = 2 \times 10^{-3} \text{ kg mol}^{-1}$. Given that $m_{H_2} = 10^{-5} \text{ kg}$ is produced in $t_1 = 1$ minute, the charge Q_1 that passed in one minute is:

$$Q_1 = \frac{(10^{-5} \text{ kg})(2)F}{2 \times 10^{-3} \text{ kg mol}^{-1}} = 10^{-2}F \text{ Coulombs}$$

Since the current I is constant, the total charge Q_2 passing through the cells during the $t_2 = 4$ minutes of operation for the second cell is exactly four times the charge passed in one minute:

$$Q_2 = It_2 = 4It_1 = 4Q_1 = 4 \times 10^{-2}F \text{ Coulombs}$$

In the second cell, aqueous $CuSO_4$ is electrolyzed, depositing solid copper (Cu) at the cathode. The corresponding half-reaction is:



For copper deposition, the valency factor is $z_{Cu} = 2$ electrons per mole of Cu . The molar mass of copper is $M_{Cu} = 63.5 \text{ g mol}^{-1} = 63.5 \times 10^{-3} \text{ kg mol}^{-1}$. Using Faraday's law again for the second cell, the mass of copper deposited m_{Cu} by the charge Q_2 is:

$$m_{Cu} = \frac{Q_2}{F} \frac{M_{Cu}}{z_{Cu}}$$

Substituting the evaluated expressions:

$$m_{Cu} = \frac{4 \times 10^{-2} F}{F} \left(\frac{63.5 \times 10^{-3} \text{ kg mol}^{-1}}{2} \right)$$

$$m_{Cu} = (4 \times 10^{-2})(31.75 \times 10^{-3}) \text{ kg} = 127 \times 10^{-5} \text{ kg}$$

Adjusting the scientific notation to find the exact mass:

$$m_{Cu} = 1.27 \times 10^{-3} \text{ kg}$$

Equivalently, this is 1.27 grams. The mass of copper deposited is:

$$1.27 \times 10^{-3} \text{ kg}$$